Robot factory worksheet

Build an offline robot factory from ordered instructions. Start with one fixed pickup fixture and one fixed destination. Pair roles: driver edits; observer predicts and checks; swap halfway. Only the instructor enables supervised hardware.

Pair names: _____ Driver first: _____ Observer first: _____

Minutes	Duration	Session
0-20	20	Meet our factory
20-45	25	Ordered instructions
45-65	20	Observations and uncertainty
65-95	30	First Python edits
95-105	10	Break
105-120	15	Teach inspect and replay demonstration
120-165	45	Explore three stations
165-210	45	Build the physical sequence
210-230	20	Improve one failure
230-240	10	Three pair demonstrations

Make a short conveyor story

Put instructions in order and repeat a small section.

Predict: _____

One change: _____

Run and observed evidence: _____

Explain and limit: _____

Evidence source (circle): PAPER / SIMULATION / PHYSICAL

Success check: Your pair can trace the order, predict a repeat count and explain one corrected sequence.

Separate the object from its report

Recognise missing and incorrect observations.

Predict: _____

One change: _____

Run and observed evidence: _____

Explain and limit: _____

Evidence source (circle): PAPER / SIMULATION / PHYSICAL

Success check: You identify one missing and one wrong report and choose pause/inspect for uncertainty.

Change a name and a number

Read and edit a short Python program before sharing the same program representation with blocks.

Predict: _____

One change: _____

Run and observed evidence: _____

Explain and limit: _____

Evidence source (circle): PAPER / SIMULATION / PHYSICAL

Success check: Both partners make one edit and explain it; shared commands are validated before execution.

Watch teach inspect return replay

Understand that a recorded stroke needs validated start/end poses and explicit jaw state.

Predict: _____

One change: _____

Run and observed evidence: _____

Explain and limit: _____

Evidence source (circle): PAPER / SIMULATION / PHYSICAL

Success check: You can say teach → inspect/validate → return to start → replay, and point to STOP.

Explore three stations

Connect observations, instructions and the robot model.

Predict: _____

One change: _____

Run and observed evidence: _____

Explain and limit: _____

Evidence source (circle): PAPER / SIMULATION / PHYSICAL

Success check: Every pair completes all three positions with one recorded prediction and evidence statement per position.

Dispense carry pick and place

Assemble the supervised fixed pickup to fixed destination sequence.

Predict: _____

One change: _____

Run and observed evidence: _____

Explain and limit: _____

Evidence source (circle): PAPER / SIMULATION / PHYSICAL

Success check: Your pair explains the full fixed-fixture sequence and records honest per-step evidence. Physical completion requires observed object transfer; offline completion is labelled rehearsal.

Find one failure and test a repair

Use a controlled change and evidence to improve a sequence.

Predict: _____

One change: _____

Run and observed evidence: _____

Explain and limit: _____

Evidence source (circle): PAPER / SIMULATION / PHYSICAL

Success check: You show a failing prediction, one repair and a repeated check with its limits.

Show one prediction and result

Explain what your program did and what the evidence cannot prove.

Predict: _____

One change: _____

Run and observed evidence: _____

Explain and limit: _____

Evidence source (circle): PAPER / SIMULATION / PHYSICAL

Success check: All three pairs demonstrate and state a specific limitation.